

Rational Points on Elliptic Curves Solutions

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1 Chapter 1

P 1.11

(a) $P + Q = \mathcal{O} * (P * Q) = \mathcal{O} * (Q * P) = Q + P$. $P + \mathcal{O} = \mathcal{O} * (P * \mathcal{O}) = \mathcal{O} * (\mathcal{O} * P) = P$.

(b)

$$\begin{aligned} X + P &= Q \\ \mathcal{O} * (X * P) &= Q \\ \mathcal{O} * (\mathcal{O} * (X * P)) &= \mathcal{O} * Q \\ X * P &= \mathcal{O} * Q \\ P * (P * X) &= P * (Q * \mathcal{O}) \\ X &= P * (Q * \mathcal{O}). \end{aligned}$$

(c)

$$\begin{aligned} (P + Q) + R &= P + (Q + R) \\ \mathcal{O} * ((P + Q) * R) &= \mathcal{O} * (P * (Q + R)) \\ (P + Q) * R &= P * (Q + R) \\ R * (P + Q) &= P * (Q + R). \end{aligned}$$

(d) If we have a surjective map $\varphi : G \rightarrow H$, where G is a group and H is a set with an operation such that $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$ for all $g_1, g_2 \in G$, then H is also a group. Take $G = H = S$.

$$\begin{aligned} (P + \mathcal{O}') +' (Q + \mathcal{O}') &= \mathcal{O}' * ((P + \mathcal{O}') * (Q + \mathcal{O}')) \\ &= \mathcal{O}' * (\mathcal{O}' * (P + Q + \mathcal{O}')) \\ &= P + Q + \mathcal{O}'. \end{aligned}$$

Now φ is surjective because $(P - \mathcal{O}') + \mathcal{O}' = P$ and is also injective because $P + \mathcal{O}' = Q + \mathcal{O}'$ implies $P = Q$.

(e) Take S to be the finite projective plane of 7 points. Let $A, B, C, D \in S$

be such that no three are collinear, $P = AB \cap CD$, $Q = AC \cap CD$, and $R = AD \cap BC$. For $P, Q \in S$, define $P * Q$ to be the third point on the line PQ if $P \neq Q$ and P otherwise. Take $\mathcal{O} = A$. Note $B * (C + D) = B * B = B$ and $C * (B + D) = C * C = C$, so $+$ is not associative.

P 1.15: (a) The inverse transformation is given by $t = \frac{\beta}{x} + \alpha$ and $u = \frac{y}{x^2}$.

Lemma: If p is a degree n polynomial and α is some constant such that the i th derivative of p evaluates to 0 at α for $i = 0, 1, 2, \dots, n$, then p is identically zero.

Proof: We can write $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$. The derivative condition gives a triangular matrix (with nonzero diagonal entries) in a_i , forcing each $a_i = 0$.

It follows that for a degree n polynomial g , $g(x) = \sum_{i=1}^n \frac{g^{(i)}(\alpha)}{i!} (x - \alpha)^i$. Take $n = 4$ and x to be t so we have

$$\begin{aligned} \frac{\beta^4}{(t - \alpha)^4} g(t) &= \frac{\beta^4}{(t - \alpha)^3} g'(\alpha) + \frac{\beta^4}{2(t - \alpha)^2} g''(\alpha) + \frac{\beta^4}{6(t - \alpha)} g'''(\alpha) + \frac{\beta^4}{24} g''''(\alpha) \\ y^2 &= \beta g'(\alpha) x^3 + \frac{\beta^2 g''(\alpha)}{2} x^2 + \frac{\beta^3 g'''(\alpha)}{6} x + \frac{\beta^4 g''''(\alpha)}{24}. \end{aligned}$$

(b) Suppose g has no duplicate roots. Then $g'(\alpha) \neq 0$, so f is a cubic. Each of the other three roots of g , say $\alpha_1, \alpha_2, \alpha_3$ map to distinct x_1, x_2, x_3 . The corresponding us (equal to 0) map to ys equal to 0. Hence f has three distinct roots, namely the x_i s. $y^2 = f(x)$ is indeed an elliptic curve.

P 1.18: (a) $Q_2 = -2Q_1 + Q_3$, $Q_4 = Q_1 - Q_3$, $Q_5 = -2Q_1$.
(b) $Q_6 = Q_3 - Q_4 = (43, 282)$. $Q_7 = Q_1 - Q_2 = (52, 375)$.
(c) $Q_8 = -Q_6 + Q_7 = -4Q_1 + 3Q_3 = (5234, 378661)$.

P 1.21: (a)

$$\begin{aligned} x(2P) &= \frac{f'(x)^2 - 4(a + 2x)f(x)}{4f(x)} \\ &= \frac{(3x^2 + 2ax + b)^2 - 4(a + 2x)(x^3 + ax^2 + bx + c)}{4x^3 + 4ax^2 + 4bx + 4c} \\ &= \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4x^3 + 4ax^2 + 4bx + 4c}. \end{aligned}$$

(b) When x is a root of f the numerator becomes $f'(x)^2$. Since f has distinct roots, f and f' share no common roots, so the numerator is nonzero.